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Detachment indicator for implant deployment

Abstract

A delivery system for deploying an implantable medical device to a target location of a body vessel can include a tubular body with a lumen extending therethrough, a pull wire extending through the lumen, and a proximal support tube disposed within the lumen and engaged to the pull wire such that proximal translation of the support tube causes proximal translation of the pull wire. The proximal support tube can include a marked distal section that is not visible to an operator of the delivery system prior to the release of the implant and that becomes visible when the proximal support tube is pulled proximally a sufficient distance to cause the delivery system to release the implant.

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Background/Summary

FIELD OF INVENTION

(1) The present invention relate to aneurysm treatment devices and more particularly, to a delivery system for deploying an implantable medical device to a target location of a body vessel that includes a detachment indicator indicating when the implantable medical device has been deployed.

BACKGROUND

(2) Numerous intravascular implant devices are known in the field. Many are deployed mechanically, via systems that combine one or more catheters and wires for delivery.

(3) Examples of implants that can be delivered mechanically include embolic elements, stents, grafts, drug delivery implants, flow diverters, filters, stimulation leads, sensing leads, or other implantable structures delivered through a microcatheter. A physician desires feedback that a

detachment cycle has been completed correctly. In most mechanical delivery systems, an inner tube is held within a lumen of a delivery system pusher and is attached to an internal wire that travels the length of the device. When pulled, the internal wire translates backwards to detach the implant. In traditional systems, an identifier that the detachment cycle has been completed is that the inner tube has been translated and thus increases in length. However, in traditional deployment systems, it is difficult for the physician to visually determine whether the implant has been deployed successfully because the small change in length is difficult to determine visually.

(4) Accordingly, there is a need for a deployment system that has a detachment indicator indicating to the physician that the implant has been successfully deployed.

SUMMARY

(5) Disclosed herein are various exemplary systems, devices, and methods of the present disclosure that can address the above needs. Examples can generally include a delivery system for deploying an implantable medical device to a target location of a body vessel. The delivery system can include a tubular body that includes a lumen extending therethrough. The delivery system can include a pull wire that extends through the lumen. The delivery system can include a proximal support tube that is disposed within the lumen. The proximal support tube can include a marked distal section that is covered by the tubular body such that the marked distal section is not visible to an operator of the delivery system. The proximal support tube can be configured to engage a proximal end of the pull wire such that proximal translation of the proximal support tube causes the pull wire to be translated proximally thereby releasing an implantable medical device from a distal end of the tubular body. The proximal support tube can be configured such that proximal translation of the proximal support tube resulting in release of the implantable medical device moves the marked distal section from the lumen of the tubular body so that the marked distal section is visible.

(6) The delivery system can further include a loop wire that includes a loop opening at a distal portion of the loop wire positioned approximate the distal end of the tubular body. The loop wire and the pull wire are positioned to secure the implantable medical device to the delivery system.

(7) The implantable medical device can be deployed in response to proximal translation of the proximal support tube, which can cause a distal end of the pull wire to exit the loop opening, which can cause the loop wire and the pull wire to disengage the medical device from the delivery system.

(8) The marked distal section can include a laser etching visible to the operator of the delivery system upon deployment of the implantable medical device.

(9) The marked distal section can include an anodized portion of the proximal support tube visible to an operator of the delivery system upon deployment of the implantable medical device.

(10) The marked distal section can include a printed or dipped portion of the proximal support tube visible to an operator of the delivery system upon deployment of the implantable medical device.

(11) The marked distal section can be visibly distinguishable from a remainder of the proximal support tube.

(12) A remainder of the proximal support tube can include a first color, and the marked distal section can include a second color distinguishable from the first color.

(13) Proximal translation of the proximal support tube can be between approximately 6 millimeters and approximately 12 millimeters to expose the marked distal portion.

(14) Proximal translation of the proximal support tube can be between approximately 6 millimeters and approximately 12 millimeters to deploy the implantable medical device.

(15) The tubular body can include a proximal tube, a flexible tube distal to the proximal tube, and a distal tube distal to the flexible tube.

(16) The flexible tube can include interference cuts.

(17) The distal tube can further include a compressed distal portion positioned approximate the implantable medical device. The compressed portion can provide an elastic force against the implantable medical device to thereby facilitate the release of the implantable medical device from

the delivery system.

(18) The compressed distal portion can include a spiral cut portion of the distal tube.

(19) In another aspect, a method for using a delivery system to deploy an implantable medical device to a target location of a body vessel is disclosed. The method can include providing a tubular body. The tubular body can include a lumen extending therethrough. The method can include extending a pull wire through the lumen. The method can include providing a proximal support tube that is disposed within the lumen. The proximal support tube can include a marked distal section that is covered by the tubular body such that the marked distal section is not visible to an operator of the delivery system. The method can include engaging a proximal end of the pull wire to the proximal support tube such that proximal translation of the proximal support tube causes the pull wire to be translated proximally. The method can include securing the implantable medical device to the tubular body with the pull wire. The method can include delivering the implantable medical device to the target location of the body vessel while the marked distal section remains covered by the tubular body. The method can include translating a proximal support tube proximally to release the implantable medical device from a distal end of the tubular body, thereby moving the marked distal section proximally out of the lumen so that the marked distal section is visible.

(20) The marked distal section can include a laser etching visible to an operator of the delivery system upon deployment of the implantable medical device.

(21) The marked distal section can include an anodized portion of the proximal support tube visible to an operator of the delivery system upon deployment of the implantable medical device.

(22) The marked distal section can include a printed or dipped portion of the proximal support tube visible to an operator of the delivery system upon deployment of the implantable medical device.

(23) A remainder of the proximal support tube can include a first color and the marked distal section can include a second color distinguishable from the first color.

(24) Proximal translation of the proximal support tube of between approximately 6 millimeters and approximately 12 millimeters exposes the marked distal portion and is effective to deploy the implantable medical device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and further aspects of this invention are further discussed with reference to the following description in conjunction with the accompanying drawings, in which like numerals indicate like structural elements and features in various figures. The drawings are not necessarily to scale, emphasis instead being placed upon illustrating principles of the invention. The figures depict one or more implementations of the inventive devices, by way of example only, not by way of limitation.

(2) FIG. 1 is an illustration of a delivery system and implant, according to aspects of the present invention.

(3) FIG. 2A-2C illustrate a deployment handle and delivery system in an unconnected state.

(4) FIG. 3 illustrates a deployment handle and delivery system in an unconnected state.

(5) FIG. 4 illustrates a deployment handle and delivery system in a connected state.

(6) FIG. 5 illustrates a deployment handle and delivery system as the deployment handle is used to deploy an implant from the delivery system.

(7) FIG. 6 shows a deployment handle and delivery system in an unconnected state after the implant is deployed.

(8) FIG. 7 is an illustration of embolic coils being positioned within an aneurysm according to aspects of the present invention.

(9) FIGS. **8A-8D** illustrate a sequence of steps for releasing an implant from the delivery system, according to aspects of the present invention.

(10) FIG. **9** is a flowchart of an exemplary method of using the deployment handle and delivery system to deploy an implant.

DETAILED DESCRIPTION

(11) The following description of certain examples of the invention should not be used to limit the scope of the present invention. The drawings, which are not necessarily to scale, depict selected embodiments and are not intended to limit the scope of the invention. The detailed description illustrates by way of example, not by way of limitation, the principles of the invention. Other examples, features, aspects, embodiments, and advantages of the invention will become apparent to those skilled in the pertinent art from the following description, which is by way of illustration, one of the best modes contemplated for carrying out the invention. As will be realized, the invention is capable of other different or equivalent aspects, all without departing from the invention.

Accordingly, the drawings and descriptions should be regarded as illustrative in nature and not restrictive.

(12) Any one or more of the teachings, expressions, versions, examples, etc. described herein may be combined with any one or more of the other teachings, expressions, versions, examples, etc. that are described herein. The following-described teachings, expressions, versions, examples, etc. should therefore not be viewed in isolation relative to each other. Various suitable ways in which the teachings herein may be combined will be readily apparent to those skilled in the pertinent art in view of the teachings herein. Such modifications and variations are intended to be included within the scope of the claims.

(13) As used herein, the terms “about” or “approximately” for any numerical values or ranges indicate a suitable dimensional tolerance that allows the part or collection of components to function for its intended purpose as described herein. More specifically, “about” or “approximately” may refer to the range of values $\pm 10\%$ of the recited value, e.g. “about 90%” may refer to the range of values from 81% to 99%. In addition, as used herein, the terms “patient,” “host,” “user,” and “subject” refer to any human or animal subject and are not intended to limit the systems or methods to human use, although use of the subject invention in a human patient represents a preferred embodiment.

(14) Turning to the figures, FIG. **1** shows an example delivery system **10** for deploying an implantable medical device **12** to a target location of a body vessel. The example delivery system **10** can include a proximal tube **100**, a flexible tube **550** positioned distal to the proximal tube **100**, and a distal tube **570** positioned distal to the flexible tube **550**. Flexible tube **550** has proximal end **552** and distal end **554**. The proximal tube **100**, flexible tube **550**, and distal tube **570** may collectively have a lumen **608** extending therethrough. That is, lumen **608** extends from a proximal end of proximal tube **100**, through flexible tube **550**, and through distal tube **570**. Collectively, proximal tube **100**, flexible tube **550** and distal tube **570** can be referred to as a tubular body **90**. The tubular body can include a proximal end **92** and a distal end **94**, as shown in FIG. **1**. Implant **12** can be positioned distal to the distal tube **570**.

(15) The flexible tube **550** can include a coiled wire, support coil that is attached at each end to the proximal tube **110** and the distal tube **570** respectively. Alternatively, the flexible tube **550** can include a laser cut hypotube including interference cuts **560** as illustrated in FIG. **1**. In this example, the flexible tube **550** is preferably contiguous with the proximal tube **100** and the distal tube **570** such that the tubes **100**, **550**, **570** are formed from a single hypotube. The interference cuts **560** can provide a high level of flexibility to flexible tube **550**. A sleeve **500** can surround the flexible tube **550** to provide a smooth outer surface of the delivery system **10** over the flexible tube **550**.

(16) An inner support tube **510** can be positioned within the lumen **608** of the flexible tube **550**. The inner support tube **510** can be welded or otherwise attached to the inner sidewall of flexible

tube **550**. Inner support tube **510** can provide structural support to the flexible tube **550**. In some examples, the inner support tube **510** can prevent the flexible tube **550** from elongating. The inner support tube **510** is optional if the interference cuts are made to allow the flexible tube **550** to have sufficient structural support in the absence of the support tube **510**, or if other features are included to provide structural support to the flexible tube **550**.

(17) A pull wire **140** can extend through the lumen **608** of delivery system **10**. Within lumen **608** of proximal tube **100** can be positioned a proximal support tube **110**. Proximal support tube **110** can be welded or otherwise attached to the inner sidewall of proximal tube **100**. Both proximal support tube **110** and inner support tube **510** can be considered as part of tubular body **90**. Proximal support tube **110** can be considered to be located on proximal end **92** of the tubular body **90**. The pull wire **140** can include a pull wire bead **142** located on a proximal end of pull wire **140**. The pull wire bead **142** can be sized such that the pull wire bead **142** is larger than the space between the proximal support tube **110** such that the pull wire bead **142** interferes with the proximal support tube **110** and proximal support tube **110** retains the pull wire bead **142** proximal to the proximal tube **100**. Alternatively, the proximal end of the pull wire can be welded to the proximal tube **110** or otherwise engaged to the proximal tube **110** as understood by a person skilled in the pertinent art.

(18) A loop wire opening **405** of loop wire **400** (also illustrated with respect to FIGS. **8A-8D**) can be located at a distal portion **404** of loop wire **400** and can extend through a locking member **18** of implant **12**. The distal end **144** of pull wire **140** can be positioned through the loop wire opening **405** of loop wire **400** to secure implant **12** to the delivery system **10**. Loop wire proximal ends **406**, **408** can be attached to the distal tube **570** as illustrated. Additionally or alternatively, one or both of the loop wire proximal ends **406**, **408** can be affixed to the proximal tube **100** to provide structural support to the flexible tube **550**. Additionally, or alternatively, the loop wire proximal ends **406**, **408** can be attached to the pull wire **140**. In this example, the loop wire **400** can be made of an elastic material, such that proximal translation of the pull wire **140** can cause the loop wire **400** to stretch and become tensioned, which provides a resistant force against further proximal translation of pull wire **140**. The loop wire **400** can be effective to prevent premature detachment of implant **12** from delivery system **10**.

(19) The proximal support tube **110** can include a marked distal section **114** that is located within the proximal tube **100**. The marked distal section **114** can be configured to be hidden from view of an operator of the delivery system **10** while implant **12** is delivered to a target location within a patient. When the pull wire **140** is pulled proximally a distance sufficient to detach implant **12**, the proximal support tube **110** can be configured to translate proximally, thereby causing the marked distal section **114** to exit the proximal tube **100** and become visible to an operator of the delivery system **10**.

(20) When the delivery system **10** is assembled, the flexible tube **550** and sleeve **500** can be more flexible than the distal tube **570** and the proximal tube **100**. One way to measure flexibility is to perform a three-point bend test wherein a portion of the delivery system **10** is held fixed at two end points, a force is applied perpendicularly to the delivery system **10** centrally between the points, and flexibility is quantified by the length of deflection of the delivery system **10** caused by the force. When measured in this way, in some examples, the flexible tube **550** and sleeve **500** can be about 1.5 times more flexible than the distal tube **570** and about 20 times more flexible than the proximal tube **100**. That is, when the three-point test is performed identically on the three sections **100**, **550**, and **570**, the flexible tube **550** can deflect over a length that is about 1.5 times the deflection of the distal tube **570** and about 20 times the length of deflection of the proximal tube **100**. Flexibility can be measured in other ways as would be appreciated and understood by a person having pertinent skill in the art.

(21) According to some embodiments, the distal tube **570** can include a compressible distal portion **300** (as shown in greater detail in FIGS. **8A** through **8D**), which can be compressed and formed from a spiral-cut portion of the distal tube **570**, formed by a laser cutting operation. Additionally, or

alternatively, the compressible distal portion **300** can be formed of a wound wire, spiral ribbon, or other arrangement allowing axial adjustment according to the present invention. Preferably, compressible distal portion **300** is in the elongated condition at rest and automatically or resiliently returns to the elongated condition from a compressed condition, unless otherwise constrained. In some embodiments, the loop wire **400** is effective to hold the compressed distal portion **300** in compression while the implant **12** is delivered to the treatment site. Alternatively, the distal tube **570** can lack the compressible spring structure **300**.

(22) Pull wire **140** can be constructed out of any suitable material, for example, pull wire **140** can be constructed of stainless steel or memory shape material, such as nitinol. According to some embodiments, pull wire **140** can additionally be coated with polytetrafluoroethylene (PTFE).

(23) FIGS. 2A through 2C are illustration of an exemplary handle member **50**, according to aspects of the present disclosure. A deployment handle **50** can include components as described in pending U.S. application Ser. No. 17/564,764, published as U.S. Patent Application Publication No. US 2023/0200819 A1 on Jun. 29, 2023, which is herein incorporated by reference as if included in its entirety. Numerous types of handle members can be used to retract the support tube **110** and pull wire **140** as understood by a person skilled in the pertinent art.

(24) FIG. 2A is a top view perspective illustration of handle member **50** from the distal end, FIG. 2B is a bottom cut-away view of handle member **50** from the proximal end, and FIG. 2C is a top view perspective illustration of handle member **50** from the proximal end. The exemplary handle member **50** can include a handle button **52**, an aperture **54**, and a handle channel **56**. The handle member may have a length L extending from a proximal end of the handle member **50** to the distal end of the handle member **50**. According to some aspects of the present disclosure, the length L of handle member **50** can be approximately between 10 cm and 20 cm.

(25) The handle channel **56**, found on the bottom of handle member **50a**, can be configured to engage with proximal support tube **110** and can have a length sufficient to engage to the length of proximal support tube **110**. Aperture **54** has a diameter sufficient to fit proximal support tube **110** therethrough. Proximal support tube **110** is configured to pass through aperture **54** when engaging to handle member **50** at handle channel **56**.

(26) Handle button **52** can be configured to be pressed by an operator of the delivery system **10**. When handle button **52** is pressed (e.g., towards a bottom surface of handle member **50**), the proximal support tube **110** can be pressed against the top surface of handle channel **56**, which can cause the support tube **110** and pull wire **140** to be retracted, thereby deploying the implant **12**. When the proximal end of the delivery system **10** is removed from the handle **50**, the marked distal section **114** of the support tube **110** can be visible to the operator. The marked distal section **114** provides a visible indication to the operator that the delivery system **10** has been properly manipulated to deploy the implant **12**.

(27) FIGS. 3-6 illustrate use of the delivery system **10** with another example handle **50**. The operator can slide handle button **52** proximally to thereby detach the implant from the deployment system **10** and expose the marked distal section **114** of the proximal support tube **110** to provide a visual indication to the operator that the implant **12** has been successfully detached from the deployment system **10** and deployed in the target location of a patient's vasculature.

(28) FIG. 3 illustrates a deployment handle **50** and delivery system **10** in an unconnected state with the handle **50**. The proximal direction (P) and distal direction (D) are indicated in the illustration. The proximal tube **100** can have a proximal support tube **110** extending in a proximal direction (P) therefrom. The proximal support tube **110** can be configured to be inserted into aperture **54** of handle member **50**. As shown, handle member **50** can include a handle button **52**, which allows the operator of the delivery system **10** and the handle member **50** to engage the proximal support tube **110** against the handle channel **56** such that the proximal support tube **110** can be translated proximally by the operator slide handle button **52** in a proximal direction (P) while handle button **52** is depressed. Proximal translation of the proximal support tube **110** can cause the pull wire **140**

to be translated proximally to thereby release the implant **12** from the delivery system **10**. As shown in FIG. **3**, the marked distal section **114** of the proximal support tube **110** is not visible, because the implant **12** has not yet been deployed from the delivery system **10**.

(29) FIG. **4** illustrates a deployment handle and delivery system in a connected state. As shown, an operator of the handle **50** and delivery system **10** can connect the proximal support tube **110** that extends from the proximal tube **100** by inserting the proximal support tube **110** into the aperture **54** of handle member **50**. Proximal support tube **110** can be inserted into aperture **54** of handle member **50** until the proximal support tube **110** extends fully through a handle channel similar to handle channel **56**, as described with respect to FIGS. **2A** through **2C**.

(30) FIG. **5** illustrates a deployment handle and delivery system as the deployment handle is used to deploy an implant **12** from the delivery system **10**. After the proximal support tube **110** is inserted through aperture **54** and fully extended into handle channel **56**, the operator can slide the handle button **52** proximally. As handle button **52** is slid proximally, the proximal support tube **110** can slide proximally. As proximal support tube **110** slides proximally, the pull wire **140** can be proximally translated with the proximal translation of the proximal support tube **110**. The proximal translation of pull wire **140** can deploy the implant **12** from delivery system **10**. In some examples, a proximal shift of between approximately 6 mm and approximately 12 mm can be effective to deploy the implant **12** from the delivery system **10**. In some examples, movement of handle button **52** proximally by between approximately 6 mm and approximately 12 mm may cause the proximal support tube **110** and the pull wire **140** to translate proximally by between approximately 6 mm and approximately 12 mm, which can cause the implant **12** to be deployed from delivery system **10**.

(31) Alternatively, the handle **50** may include a shuttle system to move the proximal support tube **110** a greater distance than the distance through which the handle button **52** travels.

(32) FIG. **6** shows a deployment handle and delivery system in an unconnected state after the implant **12** is deployed. After the proximal support tube **110** and pull wire **140** are translated proximally to deploy implant **12** from delivery system **10**, the marked distal section **114** can be exposed to an operator of the handle **50** and delivery system **10**. In some examples, the marked distal section **114** can be created by laser etching a distal portion of the proximal support tube **110** in order to create the marked distal section **114** that is visually distinct from the remainder of the proximal support tube **110**. In some examples, the marked distal section **114** can be created by an anodizing process that is applied to a distal portion of the proximal support tube **110** in order to create the marked distal section **114** that is visually distinct from the remainder of the proximal support tube **110**. In some examples, the marked distal section **114** can be created by applying a color printing process to a distal portion of the proximal support tube **110** in order to create the marked distal section **114** that is visually distinct from the remainder of the proximal support tube **110**. In some examples, the marked distal section **114** can be created by dipping a distal portion of the proximal support tube **110** in a respective paint color in order to create the marked distal section **114** that is visually distinct from the remainder of the proximal support tube **110**. In some examples, the marked distal section **114** can be a first color, while the remainder of the proximal support tube can be a second color that is visually distinct from the first color.

(33) FIG. **7** is an illustration of embolic implant **12** being delivered through catheter **750** and positioned within an aneurysm **A** on a blood vessel **BV**. The implant **12** can loop and bend with the aneurysm sac to form a thrombotic mass. The implant can loop back on themselves and/or loop next to other implants. As the aneurysm **A** becomes increasingly packed, overlapping portions of the implant **12** can press into each other. As illustrated, the implant **12** includes an embolic coil. Numerous types of implants can be deployed by a pull wire mechanism as understood by a person skilled in the pertinent art. For instance, implants can include an embolic braid, a stent, or flow diverter structure.

(34) FIGS. **8A-8D** illustrate a time sequence of steps for releasing an embolic implant **12** from a

delivery system **10**. The delivery system **10** and implant **12** can be configured such as illustrated in the previous figures and as otherwise described herein. FIG. **8A** illustrates an engagement system including the loop wire **400** and pull wire **140** locked into a locking portion **18** of the medical device **12**, thereby forming a sub-assembly. The spiral cuts **306** of the compressible distal portion **300** can be compressed and the loop wire **400** opening **405** at a distal end **404** of the loop wire **400** can be placed through the locking portion **18**. When the pull wire **140** is put through the opening **405** the medical device **12** is now secure. FIG. **8B** illustrates the pull wire **140** being drawn proximally to begin the release sequence for the medical device **12**. FIG. **8C** illustrates the instant the distal end **144** of the pull wire **140** exits the opening **405** and the pull wire **140** is pulled free of the loop wire **400**. The distal end **404** of the loop wire **400** falls away and disengaged from the locking portion **18**. As can be seen, there is now nothing holding the implant **12** to the detachment system **10**. FIG. **8D** illustrates the end of the release sequence. Here, the compressible distal portion **300** has extended/returned to its original shape and “sprung” forward. An elastic force **E** is imparted by the distal end **304** of the compressible distal portion **300** to the medical device **12** to “push” it away to ensure a clean separation and delivery of the medical device **12**.

(35) The compressible distal portion **306** can have a difference in length (distance of compression) when measured in the compressed configuration and the original, uncompressed configuration of about 0.5 mm to about 0.75 mm. Greater elastic force **E** can be achieved by using a greater distance of compression. The distance of compression can be determined by the sizing of the loop wire **400**, the shape of the locking portion **18**, and the shape of the distal end **304** of the compressible distal portion **300**.

(36) FIG. **9** is a flowchart of an exemplary method **900** of using the deployment handle and delivery system to deploy an implant. In block **904**, the method can include providing a tubular body **90** that includes a lumen **608** extending therethrough. In block **908**, the method can include extending pull wire **140** through the lumen **608**. In block **912**, the method can include providing a proximal support tube **110** that disposed within lumen **608**. The proximal support tube **110** can include a marked distal section **114** that is covered by the tubular body **90**. The marked distal section **114** can be invisible to an operator of the delivery system **10**.

(37) In block **916**, the method can include engaging a proximal end **142** of the pull wire **140** to the proximal support tube **110**. Proximal translation of the proximal support tube **110** can cause the pull **140** to be translated proximally. In block **920**, the method can include securing the implantable medical device **12** to the tubular body **90** with the pull wire **140**.

(38) In block **924**, the method can include delivering the implantable medical device **12** to the target location of the body vessel. While delivering the implantable medical device, the marked distal section **114** can remain covered by the tubular body **90** such that it is not visible to the operator of the delivery system.

(39) In block **928**, the method can include translating the proximal support tube **110** proximally to release the implantable medical device **12** from a distal end **94** of the tubular body **90**. The marked distal section **114** can be moved proximally out of the lumen **608** such that the marked distal section **114** becomes visible to an operator of the delivery system. The marked distal section can thereby effectively indicate that the implantable medical device **12** has been successfully deployed at the target location of the body vessel.

(40) The descriptions contained herein are examples of embodiments of the invention and are not intended in any way to limit the scope of the invention. As described herein, the invention contemplates many variations and modifications of the implantation system and associated methods, including alternative geometries of system components, alternative materials, additional or alternative method steps, etc. Modifications apparent to those skilled in the pertinent art are intended to be within the scope of the claims which follow.

Claims

1. A delivery system for deploying an implantable medical device to a target location of a body vessel, the delivery system comprising: a handle comprising an aperture, a handle channel, and a handle button; a tubular body comprising a lumen extending therethrough; a pull wire extending through the lumen; and a proximal support tube disposed within the lumen, a distal portion of the proximal support tube comprising a marked distal section covered by the tubular body such that the marked distal section is not visible to an operator of the delivery system, wherein the proximal support tube engages a proximal end of the pull wire such that proximal translation of the proximal support tube causes the pull wire to be translated proximally thereby releasing the implantable medical device from a distal end of the tubular body, wherein the proximal support tube is inserted into the aperture of the handle, and interacts with the handle button such that proximal movement of the handle button results in proximal translation of the proximal support tube, wherein at least a proximal portion of a length of the proximal support tube extends through the handle channel, wherein proximal translation of the proximal support tube and the handle button results in release of the implantable medical device and moves the marked distal section proximally from the lumen of the tubular body into the handle channel so that the marked distal section is visible only when the handle is completely removed from the tubular body and the proximal support tube, and wherein the handle is completely removed from the tubular body and the proximal support tube after the release of the implantable medical device.
2. The delivery system of claim 1, further comprising: a loop wire comprising a loop opening at a distal portion of the loop wire positioned approximate the distal end of the tubular body, wherein the loop wire and the pull wire are positioned to secure the implantable medical device to the delivery system.
3. The delivery system of claim 2, wherein the implantable medical device is deployed in response to proximal translation of the proximal support tube, which causes a distal end of the pull wire to exit the loop opening, which causes the loop wire and the pull wire to disengage the implantable medical device from the delivery system.
4. The delivery system of claim 1, wherein the marked distal section comprises a laser etching visible to the operator of the delivery system upon deployment of the implantable medical device upon complete removal of the handle from the tubular body and proximal support tube.
5. The delivery system of claim 1, wherein the marked distal section comprises an anodized portion of the proximal support tube visible to the operator of the delivery system upon deployment of the implantable medical device upon complete removal of the handle from the tubular body and proximal support tube.
6. The delivery system of claim 1, wherein the marked distal section comprises a printed or dipped portion of the proximal support tube visible to the operator of the delivery system upon deployment of the implantable medical device upon complete removal of the handle from the tubular body and proximal support tube.
7. The delivery system of claim 1, wherein the marked distal section is visibly distinguishable from a remainder of the proximal support tube.
8. The delivery system of claim 1, wherein a remainder of the proximal support tube comprises a first color, and the marked distal section comprises a second color distinguishable from the first color.
9. The delivery system of claim 1, wherein proximal translation of the proximal support tube of between approximately 6 millimeters and approximately 12 millimeters exposes the marked distal section upon complete removal of the handle from the tubular body and proximal support tube.
10. The delivery system of claim 1, wherein proximal translation of the proximal support tube between approximately 6 millimeters and approximately 12 millimeters is effective to deploy the

implantable medical device.

11. The delivery system of claim 1, the tubular body further comprising: a proximal tube; a flexible tube distal to the proximal tube; and a distal tube distal to the flexible tube.

12. The delivery system of claim 11, wherein the flexible tube comprises interference cuts.

13. The delivery system of claim 11, wherein: the distal tube further comprises a compressed distal portion positioned approximate the implantable medical device; and the compressed distal portion provides an elastic force against the implantable medical device thereby facilitating the release of the implantable medical device from the delivery system.

14. The delivery system of claim 13, wherein the compressed distal portion further comprises a spiral cut portion of the distal tube.

15. A method for using a delivery system to deploy an implantable medical device to a target location of a body vessel, the method comprising: providing a tubular body comprising a lumen extending therethrough; extending a pull wire through the lumen; providing a proximal support tube disposed within the lumen, the proximal support tube comprising a marked distal section covered by the tubular body such that the marked distal section is not visible to an operator of the delivery system; engaging a proximal end of the pull wire to the proximal support tube such that proximal translation of the proximal support tube causes the pull wire to be translated proximally; engaging a proximal end of the proximal support tube to a handle via an aperture in the handle; engaging at least a proximal portion of a length of the proximal support tube with a handle channel in the handle; securing the implantable medical device to the tubular body with the pull wire; delivering the implantable medical device to the target location of the body vessel while the marked distal section remains covered by the tubular body; translating the proximal support tube proximally by proximally moving a handle button on the handle to release the implantable medical device from a distal end of the tubular body, thereby moving the marked distal section proximally out of the lumen into the handle channel so that the marked distal section is visible only when the handle is completely removed from the tubular body and proximal support tube; and removing the handle completely from the tubular body and proximal support tube after the release of the implantable medical device, thereby exposing the marked distal section.

16. The method of claim 15, wherein the marked distal section comprises a laser etching visible to the operator of the delivery system upon deployment of the implantable medical device and complete removal of the handle from the tubular body and proximal support tube.

17. The method of claim 15, wherein the marked distal section comprises an anodized portion of the proximal support tube visible to the operator of the delivery system upon deployment of the implantable medical device and complete removal of the handle from the tubular body and proximal support tube.

18. The method of claim 15, wherein the marked distal section comprises a printed or dipped portion of the proximal support tube visible to the operator of the delivery system upon deployment of the implantable medical device and complete removal of the handle from the tubular body and proximal support tube.

19. The method of claim 15, wherein a remainder of the proximal support tube comprises a first color, and the marked distal section comprises a second color distinguishable from the first color.

20. The method of claim 15, wherein proximal translation of the proximal support tube of between approximately 6 millimeters and approximately 12 millimeters exposes the marked distal section upon complete removal of the handle from the tubular body and proximal support tube and is effective to deploy the implantable medical device.
